

EFFECTS OF 5 WEEKS OF HIGH-INTENSITY INTERVAL TRAINING VS. VOLUME TRAINING IN 14-YEAR-OLD SOCCER PLAYERS

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ABSTRACT

Sperlrich, B, De Marées, M, Koehler, K, Linville, J, Holmberg, H-C, and Mester, J. Effects of 5 weeks of high-intensity interval training vs. volume training in 14-year-old soccer players. *J Strength Cond Res* 25(5): 1271–1278, 2011—High-intensity interval training (HIIT) in junior and adult soccer has been shown to improve oxygen uptake ($\dot{V}O_2$) and enhance soccer performance. The main purpose of this study was to examine the short term effects of a 5-week HIIT vs. high-volume training (HVT) program in 14-year-old soccer players regarding the effects on $\dot{V}O_{2\max}$ and 1,000-m time (T_{1000}) and on sprinting and jumping performance. In a 5-week period, 19 male soccer players with a mean (*SD*) age of 13.5 ± 0.4 years performed HIIT at close to ~90% of maximal heart rate. The HVT intensity was set at 60–75% of maximal heart rate. $\dot{V}O_{2\max}$ increased significantly (7.0%) from pre to post in HIIT but not after HVT. T_{1000} decreased significantly after HIIT (~–10 vs. ~–5 seconds in HVT). Sprint performance increased significantly in both groups from pre to posttesting without any changes in jumping performance.

KEY WORDS children, endurance, exercise, jumping, oxygen uptake, sprint

INTRODUCTION

Soccer is one of the world's most popular team sports, especially among children and teenagers. Its complex nature integrates technical, tactical, and psychophysical skills. Many studies of soccer performance in children focus on technique and tactics and

ignore other conditional factors, such as endurance, strength, and speed, especially among this age group.

Because of its nature, soccer incorporates intermittent loads from high to lower intensities (4). This varying nature of the sport provokes rapid changes in oxygen uptake during training and match, averaging 70–80% of maximal oxygen uptake (12,18,21) with covered distances of about 10–12 km (4,17,21,23) at a mean intensity close to the anaerobic threshold. In this context, recent studies confirmed the importance of intensive aerobic training in soccer players (12) on improving maximal oxygen uptake, which is considered to be the most important parameter of endurance performance (3,13). In the latter study (12), the soccer players exercised twice a week for 8 weeks during preseason, executing 4×4 minutes running intervals at 90–95% of maximal heart rate with 3 minutes of rest between sets. After this training regime, maximal oxygen uptake and lactate threshold improved by 11 and 16%. Additionally, the distance covered during a match increased by 20% when the number of sprints was doubled.

In general, little is known about high-intensity interval training (HIIT) in children, especially in soccer. Children's activities are naturally characterized by spontaneous, short-term high-intensity activities (14). Studies have shown during repetitive bouts of sprints, scattered with short recovery periods, prepubertal children maintain their performance without substantial fatigue, in comparison to adults (9,14). Furthermore, repetitive intervals at high velocities close to or higher than the maximal aerobic speed, separated by short recovery periods, may elicit high oxygen consumption in children (2,10,19,22). Baquet et al. (5), for example, executed an HIIT program for 7 weeks, twice a week for 30 minutes in physical education lessons, documenting that children's maximal oxygen uptake and maximal aerobic speed could be enhanced significantly by 8.2%. Training based on these repeated HIIT programs could also improve children's anaerobic performance (short-term muscle power, strength and speed).

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25(5)/1271–1278

Journal of Strength and Conditioning Research
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Although continuous aerobic type of activity is more scientifically established as a training mode and widely researched, repeated HIIT should be considered to enhance aerobic and anaerobic performance in children's soccer. Little research has been reported regarding the influence of HIIT on aerobic and on anaerobic performance during preseason preparation in children. The preseason preparation period in youth soccer in Germany is limited to 4–6 weeks, depending on the performance level. For this reason, effective training programs such as HIIT could improve endurance performance, leaving enough time for the enhancement of other limiting skills such as technique and tactics and sprinting and strength.

The main goal of this study was to apply HIIT in a youth soccer team, to examine the short-term effects on oxygen uptake and 1,000-m running performance. An improvement in aerobic performance after HIIT was hypothesized, with no adverse effects on sprinting and jumping ability.

METHODS

Experimental Approach to the problem

The study consisted of 1 pre-post-diagnostic phase and 1 training period with 2 intervention groups to test the hypothesis (Figure 1) of whether HIIT has a greater effect on maximal oxygen uptake, 1,000-m running time, sprint, and jumping abilities compared to high volume training (HVT). In both diagnostic phases, all participants completed a $\dot{V}O_2\text{max}$ test on a treadmill, a sprint and jump test and a 1,000-m run. During the intervention, the participants exercised either according to the HIIT or HVT program. During this period, the energy expenditure of all children was recorded via lightweight multisensor devices.

Subjects

A total of 19 children took part in this study (mean $\pm$ SD: 13.5 $\pm$ 0.4 years, weight: 51.2 $\pm$ 9.1 kg, height: 160.2 $\pm$ 8.5 cm). All children were accustomed to a training workload of >4 training units per week and have been involved in soccer training and matches for at least 3 years. All children were members of a team (<14 years) within a German Premier League club. Seven players were members of a federal junior all star team. The children were divided into a training group that mainly performed high-intensity intervals (HIIT, $n = 9$, $\dot{V}O_2\text{max}$: 55.1 $\pm$ 4.9 ml·min⁻¹·kg⁻¹) and one training group with

continuous loads of endurance training (HVT, $n = 8$, $\dot{V}O_2\text{max}$: 55.3 $\pm$ 4.3 ml·min⁻¹·kg⁻¹) according to their individual maximal oxygen uptake. The personal and anthropometric data are illustrated in Table 3.

The participants and the guardians were informed about the design of the study, with special information about possible risks and benefits, and both subsequently signed an informed consent document before the start of the study. The study protocol was performed in accordance with the declaration of Helsinki and the Ethical Committee of the German Sports University in Cologne, Germany.

Procedures

Training Protocol. The intervention of both groups focused merely on the endurance part of the training session and was administered as an extension of the regular soccer-specific training. The study period was conducted directly before the beginning of the second half of the junior soccer season with 3–4 sessions per week over the 5-week winter preparation period. The training week consisted of 4 times 1–1.5 hours of practice and 1 game. During the study, all training sessions were designed in the same manner (Figure 1): The training sessions started off with a warm-up phase of 5–10 minutes, containing flexibility exercises, short submaximum sprints, and integrating game-specific actions. Thereafter, a phase of soccer-specific drills followed. Within the soccer-specific drills, either single skill practice or team tactics in small-sided games took place. Further, the focus in this phase was set on agility (twists, turns, and jumps) and also on core strength training (sit-ups and push-ups). No additional apparatus-based strength training or weight lifting was performed. Because of this training design, soccer-specific training was

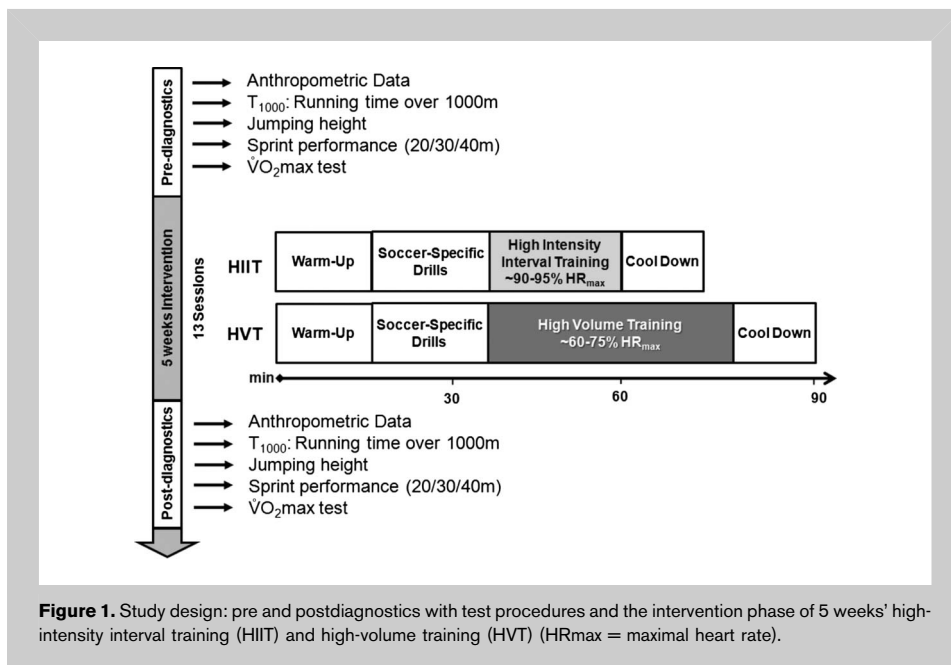


Figure 1. Study design: pre and postdiagnostics with test procedures and the intervention phase of 5 weeks' high-intensity interval training (HIIT) and high-volume training (HVT) (HR_{max} = maximal heart rate).

TABLE 1. Endurance training program and duration of each session for 5 weeks of HIIT and HVT.*

Session	HIIT		HVT	
	Program	Total T (min)	Program	Total T (min)
1	8 × 1-min + 1-min break, 6 × 1-min + 1-min break	29	6 × 6-min + 3-min break	51
2	4 × 4-min + 3-min break	29	4 × 12-min Fartlek + 2-min break	54
3	4 × 4-min + 3-min break	29	2 × 30-min Fartlek + 5-min break	65
4	12 × 30-s Sprint, 30-s break, 6 × 2-min + 2-min break	31	4 × 12-min Fartlek + 2-min break	54
5	4 × 4-min + 3-min break	29	3 × 15-min Fartlek + 3-min break	51
6	5 × 800-m + 140-s break	25	2 × 25-min Fartlek + 5-min break	55
7	10 × 400-m + 90-s break	30	Continuous run of 8.9 km	60
8	4,1,1,4,2,4-min + 2-min break	26	5 × 10-min Fartlek + 1-min break	55
9	15 × 200-m + 80-s break	29	2 × 10-min Fartlek + 3-min break, 2 × 20-min Fartlek + 3-min break	69
10	12 × 30-s sprint, 30-s break, 6 × 2-min + 2-min break	31	3 × 15-min Fartlek + 3-min break	51
11	4 × 4-min + 3-min break	29	Continuous run of 8.9 km	60
12	4 × 4-min + 3-min break	29	2 × 30-min Fartlek + 5-min break	65
13	4 × 4-min + 3-min break	29	2 × 25-min Fartlek + 5-min break	55
Mean ± SD		28.8 ± 1.7		57.3 ± 5.9

*HIIT = high intensity interval training; HVT = high volume training.

equal for both groups. After the soccer-specific part the endurance training followed. Heart rate and energy expenditure was monitored during the entire session. The high-intensity interval intervention consisted of various types of interval training without a soccer ball and did not exceed a total exercise time of 30 minutes, including rest (Table 1). During HIIT, all children should achieve or maintain 90–95% of their individual maximal heart rate, separated by periods of 1–3 minutes jogging at approximately 50–60% of maximal heart rate according to the training program (Table 1). Arterial lactate concentration and ratings of perceived exertion were obtained from every player in session 2, 7, and 13.

The HVT group performed sessions corresponding to 50–70% of their maximal heart rate without playing soccer (Figure 1). The endurance part of each session lasted 45–60 minutes.

Testing Procedure. All players within a given test procedure were tested on the same day. At first, body weight and fat free mass were measured using bioimpedance measurement (Type BF-576, Tanita, Tokyo, Japan). After these preliminary tests, all subjects completed a 20-minute warm-up at approximately 50–60% of their individual maximal oxygen uptake. Then, vertical heights of drop, squat, and counter-movement jumps were determined using a force platform with software specifically developed for the platform (Kistler, Winterthur, Switzerland). Jump height was determined as the

center of mass displacement calculated from force development and measured body mass. After this test, all participants performed a treadmill test to assess individual maximal oxygen uptake. For this, running speed was initially set to the individual running speed according to 105% of their 1,000-m personal best for 5 minutes. Thereafter, the

TABLE 2. Heart rate training zones relative to HRmax, arterial lactate concentration, and ratings of perceived exertion during 5 weeks intervention phase during HIIT and HVT.*

	HIIT	HVT	<i>p</i> †
Training intensities in % of HRmax			
>60	2.9	9.0	<0.05
60–70	13.4	22.7	<0.05
70–80	19.6	40.2	<0.05
80–90	35.8	26.0	<0.05
90–100	28.3	2.1	<0.05
Arterial lactate concentration (mmol·L ⁻¹)	8.6 ± 3.5	1.7 ± 0.7	<0.001
RPE	17.4 ± 0.9	12.2 ± 0.6	<0.001

*HIIT = high-intensity interval training; HVT = high-volume training; HRmax = maximum heart rate; RPE = ratings of perceived exertion.

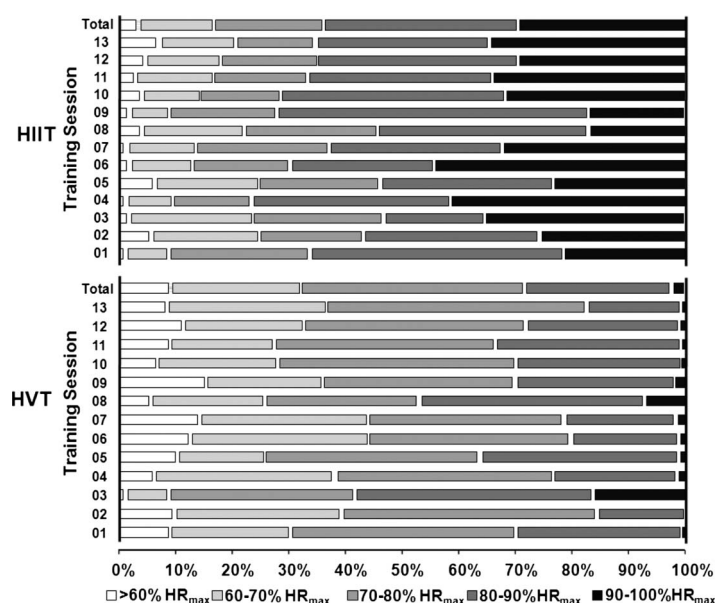


Figure 2. Training load in regard to heart rate training zones (relative to HR_{max}) during 5 weeks' high-intensity interval training (HIIT) and high-volume training (HVT).

inclination of the treadmill was increased 1° each minute to volitional exhaustion.

Oxygen uptake was measured with an open circuit breath-by-breath spirometer (nSpire, Zano600USB, Oberthurba,

Germany) throughout the testing, using standard algorithms with dynamic account for the time delay between the gas consumption and volume signal. The spirometer was calibrated before each test using calibration gas (15.8% O₂, 5% O₂ in N₂; Praxair, Germany) targeting the range of anticipated fractional gas concentration administered with a precision 1-L syringe (nSpire). Heart rate was recorded in real time every 5 seconds during the testing using short-range telemetry (Polar S 710, Kempele, Finland). All respiratory and heart rate data were averaged every 30 seconds. The highest value for oxygen uptake and heart rate within the last 30 seconds of the test was used for statistical analysis.

A 20-, 30-, and 40-m sprint test, and a 1,000-m run, followed the laboratory testing. The time for all sprint tests was measured using photocells (Brower Timing Systems, South Draper, UT, USA) at the start line, and at 20, 30, and 40m. Each subject had 2 trials separated by 5 minutes of rest. When

TABLE 3. Pre-post values for HIIT and HVT*

	HIIT			HVT		
	Pre	Post	<i>p, d</i>	Pre	post	<i>p, d</i>
1,000-m time (min:s)	03:54.4 ± 12.8	03:44.6 ± 13.8*	<0.001, 0.72	03:55.3 ± 12.8	03:50.7 ± 17.3	0.12, 0.31
$\dot{V}O_{2max}$ (ml·min ⁻¹ ·kg ⁻¹)	55.1 ± 4.9	58.9 ± 4.7*	<0.001, 0.79	55.3 ± 4.3	56.4 ± 3.7	0.21, 0.37
Fat-free mass (kg)	48.8 ± 9.8	49.8 ± 9.7	0.20, 0.10	42.5 ± 8.7	43.4 ± 8.4	0.20, 0.1
20-m Sprint time (s)	3.49 ± 0.16	3.34 ± 0.19*	0.01, 1.03	3.43 ± 0.20	3.31 ± 0.13*	0.007, 0.71
30-m Sprint time (s)	5.00 ± 0.25	4.78 ± 0.26*	0.001, 0.86	4.87 ± 0.29	4.70 ± 0.25*	0.001, 0.62
40-m Sprint time (s)	6.41 ± 0.37	6.23 ± 0.39*	0.003, 0.47	6.33 ± 0.43	6.12 ± 0.36*	<0.001, 0.53
Drop jump (m)	0.27 ± 0.05	0.31 ± 0.07	0.70, 0.66	0.31 ± 0.09	0.33 ± 0.05	0.66, 0.27
Squat jump (m)	0.28 ± 0.07	0.31 ± 0.07	0.93, 0.43	0.29 ± 0.05	0.33 ± 0.09	0.29, 0.54
Countermovement jump (m)	0.26 ± 0.05	0.29 ± 0.06	0.11, 0.54	0.27 ± 0.04	0.34 ± 0.07	0.10, 1.22

*HIIT = high intensity interval training; HVT = high volume training.

ready to sprint, the player self-started the sprint test from a static upright position. The time was recorded when the subjects broke the photocell beam.

The 1,000-m run test was performed on the soccer field in a 250-m rectangle. Time was recorded manually with a stop watch (Model "Marathon," Timex, Middlebury, CT, USA). All participants performed 3 1,000-m runs before the study to get accustomed to the length and intensity of the test.

The heart rates of all children and all training sessions during the intervention period were recorded telemetrically using a heart rate belt system for team analysis (Acentas, Hörgertshausen, Germany). All data were saved and analyzed afterward using software provided by the manufacturer. The maximal heart rate from the laboratory ramp test was used as reference value for training analysis. In 3 training sessions, a 20- μ l blood sample from the right ear lobe was collected (Eppendorf, Hamburg, Germany) and analyzed amperometric-enzymatically for blood lactate concentration using Ebio Plus (Eppendorf). Within this session the subjects were asked to rate their perceived exertion on Borg's scale. The SenseWear Pro 3 Armband (BodyMedia, Pittsburgh, PA, USA) was used to continuously assess energy expenditure 24 hours in a 1-week period, including training and match. The armband was worn on the subjects' upper arm and recorded biaxial acceleration, heat production and galvanic skin response. Energy expenditure was calculated using InnerView software (version 6.1) and is expressed as metabolic equivalents, which are multiples of the basal metabolic rate (1).

Statistical Analyses

All results are documented as mean $\pm$ SD. A Student's test was performed to analyze the differences in heart rate and lactate concentration between HIIT and HVT during the intervention. The significance of within and between-condition mean differences was assessed by analysis of variance, with repeated measures followed by post hoc analyses using the Least Significant Difference test. An alpha level of $p \leq 0.05$ was considered statistically significant and marked as $*p \leq 0.05$. Standardized difference (d) was calculated to standardize comparisons between the pre and postmeasurements. This effect size was considered small when $d < 0.2$, moderate when $d < 0.4$ and high when $d = 0.6$ or greater. The Statistica (version 7.1;

StatSoft Inc.) software package for Windows was used for all statistical analysis.

RESULTS

Intervention

Ninety-four percent of all training sessions were completed in both groups. Table 2 and Figure 1 show the percentage amount of training performed at different heart rate intervals. The HIIT showed a significantly greater amount of time spent at intensities of 80–100% of maximal heart rate compared to HVT ($p < 0.05$). HVT, on the other hand, revealed a higher percentage in the lower heart rate zones (60–80% of maximal heart rate) compared to HIIT ($p < 0.05$).

The ratings of perceived exertion in HIIT were significantly higher (17.4 ± 1.5) compared to HVT (12.2 ± 0.6 ; $p < 0.001$). Also, arterial lactate concentration was recorded at significantly higher levels during HIIT compared to HVT sessions (8.6 ± 3.5 vs. 1.7 ± 0.7 mmol·L⁻¹, $p < 0.001$).

No differences were found in energy expenditure per session and per week between the groups ($p < 0.74$).

Pre-Post Diagnostic

Body weight, height, and fat-free mass remained unaltered in both groups from pre to postintervention (Table 3). Relative oxygen uptake increased significantly by 7.0% from pre to post in HIIT but not in HVT (+1.9%). Running time over 1,000 m decreased significantly in HIIT but not in HVT (Figure 2). The mean decrease in HIIT was 10 vs. 5 seconds in HVT. Sprint performance increased significantly in both

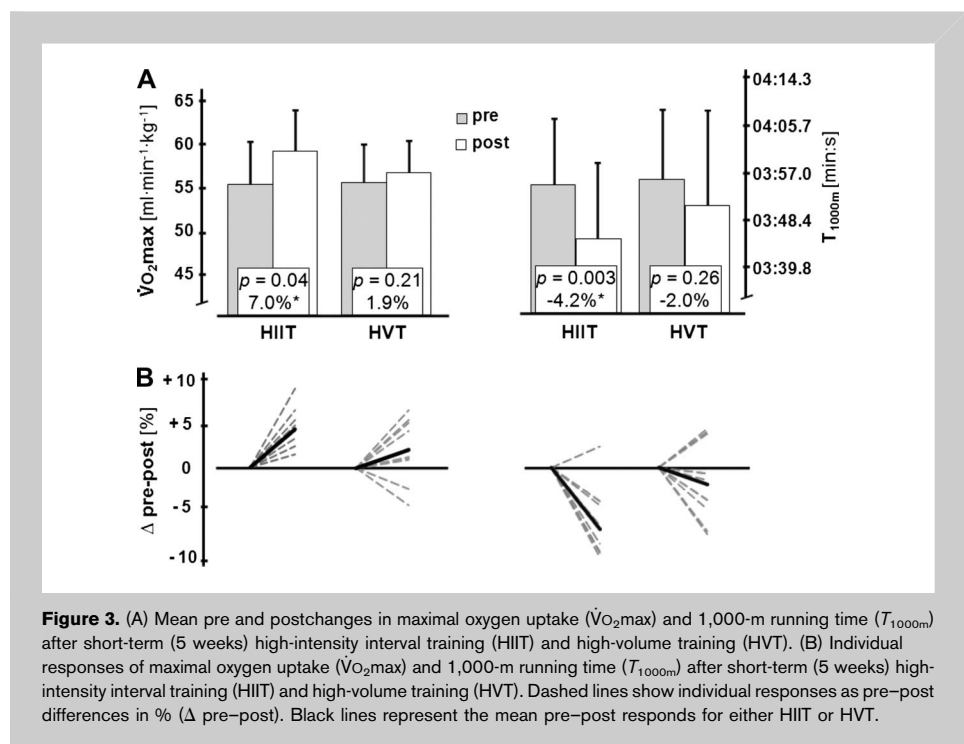


Figure 3. (A) Mean pre and postchanges in maximal oxygen uptake ($\dot{V}O_{2\max}$) and 1,000-m running time (T_{1000m}) after short-term (5 weeks) high-intensity interval training (HIIT) and high-volume training (HVT). (B) Individual responses of maximal oxygen uptake ($\dot{V}O_{2\max}$) and 1,000-m running time (T_{1000m}) after short-term (5 weeks) high-intensity interval training (HIIT) and high-volume training (HVT). Dashed lines show individual responses as pre-post differences in % (Δ pre-post). Black lines represent the mean pre-post responds for either HIIT or HVT.

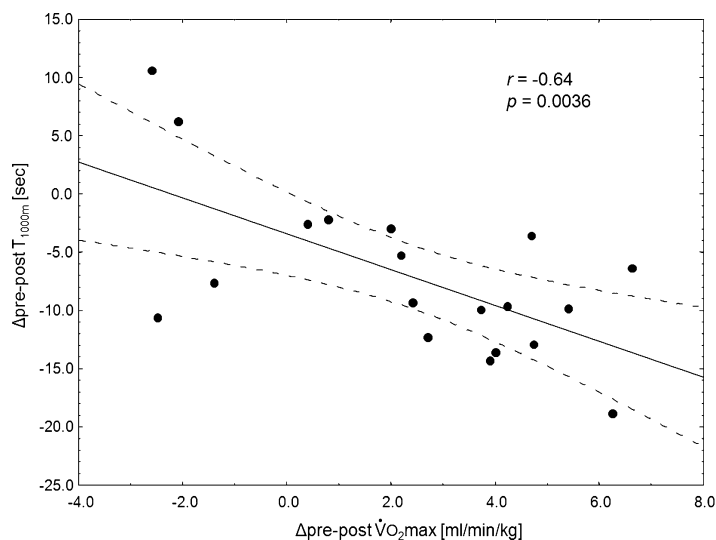


Figure 4. Correlation of pre-post changes of maximal oxygen uptake ($\Delta\text{pre-post } \dot{V}\text{O}_2\text{max}$) vs. pre-post changes in 1,000-m running time ($\Delta\text{pre-post } T_{1000\text{m}}$).

groups from pre to posttesting (Table 3) without any change in jump performance.

DISCUSSION

The overall findings revealed a significant increase in oxygen uptake of 7% and a decrease in 1,000-m running time high-intensity interval intervention. Further, sprint performance increased in both groups with no changes in jumping performance. These short-term effects were achieved after 5 weeks of HIIT vs. HVT in 14-year-old soccer players.

The applied training interventions in this study elevated mean relative oxygen uptake by $3.9 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ or 7.0% after HIIT and by $1.1 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ or 1.9% after HVT. The increase in oxygen uptake for HIIT lies within the range of previously published data for adolescent players. Macmillan et al. found an increase of 9% in oxygen uptake after HIIT in 16.9 ± 0.4 -year-old players (15). Helgerud et al. executed 4-minute intervals at 90–95% maximal heart rate with youth soccer players (Age: 18.1 ± 0.8 years) over an 8-week training period and promoted an increase in $\dot{V}\text{O}_2\text{max}$ of 11%, equivalent to an increase from 58.1 ± 4.5 to $64.3 \pm 3.9 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ (12).

Baquet et al. analyzed the effects of a 7-week interval-training program in 8- to 11-year old boys and girls exercising 2 times per week for 30 minutes using short high-intensity intermittent running (10×10 seconds or 5×20 seconds) at 80–95% HRmax (5). After training, the experimental group demonstrated a significant improvement in $\dot{V}\text{O}_2\text{max}$, from 43.9 to $47.5 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ (+8.2%). The training intervention revealed improvements of 0.58, 0.67, and 0.56% in $\dot{V}\text{O}_2\text{max}$

per training session (5,12,15), compared to 0.47% in this study. The somewhat greater efficiency of their studies compared to our intervention may be because of the longer intervention phase of 7–10 weeks compared to 5 weeks in this study. Furthermore, the children in Baquet's study demonstrated lower baseline values for oxygen uptake, which in turn favors a greater increase in aerobic capacity (5).

The mean increase in oxygen uptake of 7.0% after HIIT and 1.9% after HVT is attenuated by a mean decrease in running time over 1,000 m of 4.2% after HIIT ($\Delta -10$ second) and 2% after HVT ($\Delta -5$ second). This demonstrates a greater effect in the HIIT group compared to HVT. In this study, the increase

in maximal oxygen uptake correlates with an increase in running time ($r = -0.63$, $p = 0.0036$) indicating a correlation between the laboratory testing and simple field testing (Figure 4). An increase in oxygen uptake by $3\text{--}5 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ in this study revealed an increase in 1,000-m running time of approximately 10–15 seconds (Figure 3). So from a practical point of view, to avoid cost-worthy and complex laboratory procedures, the simple assessment of $T_{1000\text{m}}$ reflects a feasible and uncomplicated method to detect improvements in endurance performance. Further, the frequent measurement of 1,000-m time provoked a positive competition mentality among the players wanting to beat their personal best and their teammates' times.

Potential peripheral and central adaptations after HIIT vs. HVT are presented in several studies. Massicote and McNab matched frequency and duration but varied intensities by using heart rates of 130–140, 150–160, 170–180 $\text{b}\cdot\text{min}^{-1}$ in a group of boys (14). All groups increased submaximum responses expressed in work capacity but only the group exercising at highest intensities significantly increased their $\dot{V}\text{O}_2\text{max}$. Helgerud et al. (12) showed that HIIT resulted in a significantly increased $\dot{V}\text{O}_2\text{max}$ compared to long slow distance and lactate-threshold training intensities. The percentage increases for a 15-second interval (15-second rest) and 4-minute interval (4-minute rest) were 5.5 and 7.2%, respectively, demonstrating increases in maximal oxygen uptake from 60.5 to 64.4 and 55.5 to 60.4 $\text{ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$. These authors attribute the increases in maximal oxygen uptake after high-intensity training to increased stroke volume resulting in increased cardiac output. Further, fluctuations in workload and oxygen uptake during training

sessions, rather than exercise duration or global energy expenditure, are key factors in improving muscle oxidative capacities (7). Gibala et al., for example, revealed similar increases in muscle oxidative capacity after HIIT or HVT (11). This is reflected by the maximal activity of cytochrome *c* oxidase and subunits content when performing HIIT (4–6 × 30 seconds, overall training time: 18–27 minutes) when compared to traditional high-volume intensity (90–120 at 65% of maximal oxygen uptake). Burgomaster et al. (6) showed that both protocols induce similar increases in mitochondrial markers and lipid oxidation. Glycogen and phosphocreatine utilization during exercise were reduced after training, and calculated rates of whole-body carbohydrates decreased, whereas lipid oxidation increased with no differences between groups. From this point, intermittent training improves both central and peripheral components of maximal oxygen uptake, whereas continuous training is mainly associated with greater oxygen extraction (7).

A sudden increase in volume or exercise intensity over a longer period of time, for example, in intensity or volume blocks, comparable to the present intervention, could lead to so called “overreaching” or even overtraining symptoms (5,20). Such unwanted conditions may manifest in reduced maximum physical capacity (5) or “burnout” symptoms such as a washed-out feeling, tiredness, lack of energy, muscle and joint pain, and decreased immunity. None of the players or the parents reported any of these symptoms. Also, we did not notice any increased incidence of musculoskeletal injuries. Therefore, based on our observation, we could not see any of these unwanted negative effects after both training strategies.

Several authors point out that strength-related power output is impaired when executing endurance training (8,9,16). This study showed no negative effect on vertical height in drop, squat, and countermovement jumps. Therefore, it appears that power-related performance was not hindered by the HIIT nor the HVT regimes used in this study. These data are confirmed by results from Helgerud et al., showing a substantial increase in maximal oxygen uptake after 8 weeks of endurance training resulting in no reduction in sprinting and jumping abilities (12). Both groups in this study performed the same amount of soccer-specific exercise during the intervention period with an equal amount of soccer-specific sprints, twists, turns, and jumps. As these data show, the soccer-specific block in each training session seemed to have been a sufficient stimulus to increase sprint performance over 20, 30, and 40 m in both groups. The data further show that jumping height was not negatively influenced either by the type of endurance training (HIIT vs. HVT) nor by the soccer-specific training blocks. From a practical point of view, as long as there is an appropriate amount of soccer specific stimuli, it may be concluded that neither HIIT nor HVT shows negative effects on sprinting and jumping performance in 14-year-old players. Further, soccer-

specific exercise is sufficient to enhance sprinting performance, however, to improve jumping performance additional power-related exercise must be administered.

PRACTICAL APPLICATIONS

This study revealed a significant increase in oxygen uptake of 7% and a decrease in 1,000-m running time after 5 weeks of HIIT. High volume training on the other hand showed no effects on maximal oxygen uptake and 1,000-m time in 14-year-old soccer players. These short-term effects were achieved in 1.5–2.0 h·wk⁻¹ less exercise time.

From a coach's point of view, exercise time, especially in children, is limited because of other factors such as school and recreational activities. So, time saving strategies in soccer are necessary to enhance endurance-relevant parameters, such as oxygen uptake and running performance. These data suggest that short-term HIIT, performed 3× per week for 5 weeks, leads to a clear increase in relevant endurance values (maximal oxygen uptake and 1,000-m running time). Secondly, the data show an increase in sprint performance after HIIT and HVT, which is more affected because of soccer-specific drills than to the type of endurance training. And thirdly, jumping performance remains unaltered after 5 weeks of HIIT compared to HVT. As a result, power-related exercise must be administered to additionally improve jumping performance. During this time, no signs of overtraining symptoms or increased injuries were reported.

Therefore, a training period of 5 weeks' HIIT should be considered as a valuable training method in children's soccer when training time is limited. However, data on long-term effects in children are not available and should be administered carefully because of the potential risk of overtraining or injury.

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